

# CLASS -10 (2025-26)

## Values and Types

### CHAPTER 3

#### Assignments:-

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A. Tick (✓) the correct answers.

1. To store character values, a corresponding numeric value is generated in

- a. Decimal form
- b. Binary form
- c. ASCII
- d. UNICODE

Answer:- c. ASCII

2. How many escape sequence characters are there in Java?

- a. 7
- b. 10
- c. 8
- d. 12

Answer:- c. 8

3. Consider the following code:

```
datatype variable = (datatype) variable_to_be_converted;
```

Which of the following is represented in the preceding code?

- a. Type conversion
- b. Initialization
- c. Declaration
- d. Operation

Answer:- a. Type conversion

4. Which of the following is an example of a reference data type?

- a. int
- b. array
- c. float
- d. double

Answer:- b. array

5. Which of the following can be used while creating an identifier?

- a. Letters
- b. Underscore (\_)
- c. Dollar (\$)
- d. All of these

Answer:- d. All of these

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#### B. Fill in the blanks.

1. Character set in Java consists of **letters, digits, and special characters**.

Answer:- letters, digits, and special characters

2. Implicit type conversion takes place when the two types are **compatible**.

Answer:- compatible

3. Non-primitive data types are also called **reference** data types.

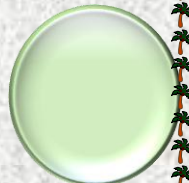
Answer:- reference

4. The size of the "short" data type is **less** than the "long" data type.

Answer:- less

5. **null** is a special Java literal which represents a null value.

Answer:- null



## C. Short Answer Type Questions

### Define String literals and boolean literals.

**Answer:-**

- **String literals** are sequences of characters enclosed in double quotes, such as "Hello".
- **Boolean literals** represent truth values in Java and can only be `true` or `false`.

### Write the difference between declaration and initialization.

**Answer:-**

- **Declaration** is when a variable is defined with a type but not necessarily given a value (e.g., `int x;`).
- **Initialization** is when a declared variable is assigned a value (e.g., `x = 5;`).

### 3. Name the various types of tokens used in Java.

**Answer:-**

The different types of tokens in Java are:

- I. Keywords
- II. Identifiers
- III. Literals
- IV. Operators
- V. Separators

### 4. What is the use of "\n" in Java?

**Answer:-**

\n is an escape sequence used in Java to move the output cursor to the next line. It is used to break lines in console output.

### What is the difference between variables and identifiers?

**Answer:-**

- A **variable** is a named location in memory used to store data during program execution.
- An **identifier** is the name given to elements like variables, classes, methods, etc. So, all variables are identifiers, but not all identifiers are variables.

## Assertion and Reason Questions with Options

1.

**Assertion (A):** Java uses UNICODE to represent character values.

**Reason (R):** UNICODE can represent characters from multiple languages around the world.

**Options:**

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true, but R is false
- d) A is false, but R is true

**Answer:-** a) Both A and R are true, and R is the correct explanation of A

2.

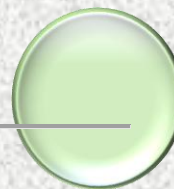
**Assertion (A):** Escape sequences in Java help in formatting output.

**Reason (R):** \n adds a new line, and \t adds a tab space in output.

**Options:**

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true, but R is false
- d) A is false, but R is true

**Answer:-** a) Both A and R are true, and R is the correct explanation of A





3.

**Assertion (A):** The expression `(datatype) variable_to_be_converted` is an example of type casting.

**Reason (R):** Type casting allows conversion of one data type into another in Java.

**Options:**

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true, but R is false
- d) A is false, but R is true

**Answer:-** a) Both A and R are true, and R is the correct explanation of A

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4.

**Assertion (A):** Arrays are primitive data types in Java.

**Reason (R):** Arrays store a fixed number of elements of the same data type.

**Options:**

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is false, but R is true
- d) A is true, but R is false

**Answer:-** c) A is false, but R is true

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5.

**Assertion (A):** Identifiers in Java can start with a digit.

**Reason (R):** Identifiers may only contain letters, digits, underscores, and dollar signs.

**Options:**

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is false, but R is true
- d) A is true, but R is false

**Answer:-** c) A is false, but R is true

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6.

**Assertion (A):** Implicit type conversion happens when types are incompatible.

**Reason (R):** Java automatically converts smaller types to larger compatible types.

**Options:**

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is false, but R is true
- d) A is true, but R is false

**Answer:-** c) A is false, but R is true

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7.

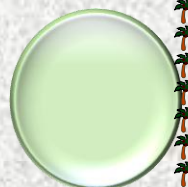
**Assertion (A):** The size of the `short` data type is greater than the `long` data type.

**Reason (R):** `long` takes 8 bytes whereas `short` takes only 2 bytes in Java.

**Options:**

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is false, but R is true
- d) A is true, but R is false

**Answer:-** c) A is false, but R is true



8.

**Assertion (A):** `null` is a special keyword used for primitive data types.

**Reason (R):** `null` represents a value that points to no object in Java.

**Options:**

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is false, but R is true
- d) A is true, but R is false

**Answer:-** c) A is false, but R is true

9.

**Assertion (A):** Declaration and initialization in Java are the same thing.

**Reason (R):** Declaration reserves memory, while initialization assigns a value to that memory.

**Options:**

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is false, but R is true
- d) A is true, but R is false

**Answer:-** c) A is false, but R is true

10.

**Assertion (A):** In Java, variables and identifiers are different concepts.

**Reason (R):** Identifiers are names given to various program elements, and variables are just one of them.

**Options:**

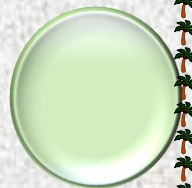
- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true, but R is false
- d) A is false, but R is true

**Answer:-** a) Both A and R are true, and R is the correct explanation of A

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## One-word questions and answers

- 1. **Q:** Character encoding used in Java?  
**A:** UNICODE
- 2. **Q:** Special literal representing no value?  
**A:** null
- 3. **Q:** Type of conversion done automatically by Java?  
**A:** Implicit
- 4. **Q:** Data type category of array in Java?  
**A:** Reference
- 5. **Q:** Symbol for new line in Java?  
**A:** \n
- 6. **Q:** Category of data types like `int`, `char`, `float`?  
**A:** Primitive
- 7. **Q:** What starts variable names in Java?  
**A:** Letter
- 8. **Q:** Token type for `+`, `-`, `*`, `/` in Java?  
**A:** Operator





9. **Q:** Data type that uses 8 bytes in Java?  
**A:** long
10. **Q:** Grouping of keywords, identifiers, literals, etc.?  
**A:** Tokens
11. **Q:** What symbol is used for tab space in Java?  
**A:** \t
12. **Q:** What is the smallest integer data type in Java?  
**A:** byte
13. **Q:** What keyword is used to define a constant in Java?  
**A:** final
14. **Q:** What type of data type is String in Java?  
**A:** Reference
15. **Q:** What is the process of converting one data type to another manually?  
**A:** Casting
16. **Q:** Which data type holds true or false values?  
**A:** boolean
17. **Q:** Which escape character is used for backspace?  
**A:** \b
18. **Q:** What keyword is used to declare a variable in Java?  
**A:** int (or any other datatype like float, char, etc.)
19. **Q:** What do you call a name used to represent variables, methods, or classes?  
**A:** Identifier
20. **Q:** What is the default value of an uninitialized reference variable?  
**A:** null

## Fill in the Blanks (with Answers)

1. Java uses UNICODE to store character values.  
**Answer:** UNICODE
2. The special literal that represents a null value in Java is null.  
**Answer:** null
3. Implicit type conversion is automatically performed by Java when compatible types are used.  
**Answer:** Implicit
4. Arrays are examples of reference data types in Java.  
**Answer:** reference
5. The escape character \n is used to insert a new line.  
**Answer:** new line
6. Java character set includes letters, digits, and special characters.  
**Answer:** special characters
7. Tokens are the smallest meaningful elements in a Java program.  
**Answer:** Tokens
8. A variable name in Java must begin with a letter, underscore, or dollar sign.  
**Answer:** letter
9. Java keywords like if, class, and while are reserved words.  
**Answer:** words
10. The keyword used to declare a constant value is final.  
**Answer:** final
11. The default value of an uninitialized reference variable is null.  
**Answer:** null
12. The process of manually converting one data type to another is called casting.  
**Answer:** casting

13. The escape character \t is used to insert a tab space.

**Answer:** tab space

14. A data type that holds true or false values is called boolean.

**Answer:** boolean

15. Byte is the smallest integer data type in Java.

**Answer:** byte

16. A **primitive** data type is predefined by the language and is not an object.

**Answer:** primitive

17. Identifiers are the names given to variables, methods, classes, etc.

**Answer:** names

18. The data type that uses 8 bytes of memory is long.

**Answer:** long

19. The escape character used for backspace is \b.

**Answer:** \b

20. A reference data type refers to an object or array in memory.

**Answer:** reference